A

Major Project

On

AI – LEND SENSE: PREDICTIVE AND EXPLAINABLE LOAN DECISION SYSTEM

(Submitted in partial fulfilment of the requirements for the award of Degree)

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In

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

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The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

"AI-Lend Sense: A Machine Learning-Based Approach for Predictive and Explainable Loan Decision Systems." The rapid growth of digital banking and financial services has significantly increased the number of loan applications processed by financial institutions. However, this expansion has also introduced challenges such as manual evaluation delays, inconsistent decision-making, and potential bias in loan approval processes. Traditional methods rely heavily on human judgment and predefined rules, which are often time-consuming, error-prone, and lack transparency for applicants. Therefore, there is a strong need for an automated, accurate, and explainable system that can assist financial institutions in making reliable loan decisions in real time. In response to these challenges, an intelligent machine learning-based framework is proposed for predicting loan approval and assessing credit risk. The system utilizes multiple supervised learning algorithms, including Logistic Regression, Decision Tree, and Random Forest, to analyze applicant information such as income, employment status, credit history, and debt-to-income ratio. These models learn patterns from historical loan data and generate predictions indicating whether a loan should be approved or rejected. To enhance transparency and trust, an Explainable Artificial Intelligence (XAI) technique, specifically LIME (Local Interpretable Model-Agnostic Explanations), is integrated into the system to provide clear, human-understandable explanations for each decision, highlighting the key factors influencing the outcome The proposed system is implemented as a user-friendly web application that allows applicants to input their details and instantly receive loan predictions along with detailed explanations. Experimental results demonstrate that ensemble-based models, particularly Random Forest, achieve high prediction accuracy compared to traditional methods, while also reducing processing time and minimizing human bias Furthermore, the system improves fairness by incorporating data preprocessing techniques and multi-model decision strategies, ensuring consistent and unbiased results. Overall, the integration of machine learning models with explainable AI techniques provides a robust, transparent, and efficient solution for automated loan decision-making. The findings indicate that combining predictive modelling with interpretability enhances trust, accuracy, and usability, making the system suitable for modern financial applications and improving both customer experience and institutional decision processes.

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1. INTRODUCTION

1. INTRODUCTION

"AI-Based Loan Decision System Using Transaction and Financial Data," is designed to develop an intelligent and automated system capable of predicting loan approval decisions accurately. The system analyzes applicant information such as income, employment status, credit history, and transaction behavior to determine whether a loan should be approved or rejected. To achieve this, the system leverages Machine Learning models such as Logistic Regression, Decision Tree, and Random Forest, ensuring reliable and data-driven decision-making.

Given the increasing number of loan applications in banks and financial institutions, manual evaluation is time-consuming and often inconsistent. Therefore, an automated AI-based system becomes essential to improve efficiency and accuracy. This project ensures that loan decisions are made quickly with minimal human intervention and reduced bias. The system is designed for scalability, enabling it to process large volumes of applications efficiently. Additionally, the project explores real-time loan decision support, making it suitable for banking systems, financial services, and online lending platforms.

1.1 PROJECT PURPOSE

The primary purpose of this project is to improve the loan approval process by automating the prediction of loan eligibility using Artificial Intelligence. Traditional loan approval systems rely heavily on manual verification, which can be slow, error-prone, and sometimes biased. By implementing machine learning techniques, this project provides a faster, more consistent, and fair method for evaluating loan applications.

With the growing demand for quick financial services, there is a strong need for systems that can instantly analyze applicant data and provide accurate decisions. By identifying patterns in historical loan data, this system predicts the probability of loan approval or default, helping financial institutions reduce risk.

By analyzing factors like income, credit score, job stability, and transaction history, the system reduces manual work, speeds up decisions, and improves transparency. Explainable AI also helps applicants understand approval or rejection, increasing trust and fairness.

1.2 PROJECT FEATURES

This project incorporates several key features to improve the accuracy, transparency, and efficiency of loan decision-making:

Automated Loan Prediction: The system uses Machine Learning algorithms such as Logistic Regression, Decision Tree, and Random Forest to analyze applicant data and predict loan approval status. These models learn from historical data and identify patterns that help in making accurate predictions, reducing the chances of human error.

Explainable Decision Making: Unlike traditional systems, this project integrates Explainable AI techniques such as LIME to provide clear explanations for each decision. It highlights important factors like low income, high debt, or poor credit score, helping both banks and applicants understand the reasoning behind the decision.

Scalability & Real-Time Processing: The system is designed to handle large datasets and provide instant results, making it suitable for real-time loan approval systems. By using optimized algorithms and efficient data processing, the system can analyze multiple applications simultaneously, ensuring faster and reliable performance in large-scale financial environments.

By integrating these features, the proposed system provides an efficient, transparent, and scalable solution for loan decision prediction, ensuring accurate risk assessment, reduced processing time, and improved customer trust in financial services.

2. LITERATURE SURVEY

2. LITERATURE SURVEY

With the rapid growth of digital banking and online financial services, loan approval systems have become a crucial component of modern financial institutions. Traditional loan approval processes are time-consuming, require manual intervention, and are prone to human bias and inconsistency. With the increasing volume of loan applications, there is a need for automated, intelligent, and scalable solutions that can accurately predict credit risk and assist in decision-making. Machine learning and deep learning techniques have been widely adopted to improve the accuracy and efficiency of loan approval systems.

This literature survey reviews recent advancements in loan decision prediction, focusing on machine learning models, deep learning architectures, hybrid approaches, explainable AI techniques, and ensemble learning methods such as Random Forest. The aim is to highlight existing approaches, their strengths and limitations, and the potential for improving loan decision systems.

Recent research in 2025 has focused on integrating Explainable AI (XAI) with deep learning models to improve transparency in loan approval systems. Researchers have emphasized the importance of interpretable models to ensure fairness and regulatory compliance in financial decision-making. Advanced XAI techniques such as SHAP and LIME are being used to explain predictions made by complex models, enabling banks and financial institutions to justify loan approval or rejection decisions more effectively. In addition, 2025 studies have explored the use of transformer-based architectures for financial prediction. Transformers, which are highly effective in handling sequential and long-range dependencies, have shown improved performance over traditional RNN and LSTM models in analyzing large-scale financial transaction data. These models help capture complex customer behavior patterns more efficiently and improve prediction accuracy.

In 2024, researchers focused on AutoML (Automated Machine Learning) and hybrid ensemble frameworks for loan prediction. AutoML techniques automate model selection, feature engineering, and hyperparameter tuning, reducing human effort and improving model performance. Studies have shown that combining AutoML with ensemble learning methods such as Random Forest and Gradient Boosting results in highly optimized and scalable loan approval systems. Another important advancement in 2024 is the use of Graph Neural

Networks (GNNs) for credit risk assessment. These models analyze relationships between customers, transactions, and financial entities, helping identify hidden risk patterns and fraudulent activities more effectively than traditional machine learning methods.

Recent studies in 2023 have highlighted the advantages of integrating ensemble learning methods, such as Random Forest and Gradient Boosting, with deep learning models. Ensemble techniques improve prediction stability and reduce overfitting by combining multiple models. Sharma et al. (2023) incorporated Random Forest into a neural network-based system, demonstrating improved robustness and reliability in loan approval decisions. The ensemble model utilized features extracted from deep learning models to refine final predictions, resulting in better generalization. Another emerging area of research in 2023 is multimodal data analysis in loan prediction systems. By combining structured financial data, transaction history, and textual data such as customer feedback, models can achieve better performance. Gupta et al. (2023) combined deep learning models with natural language processing techniques to analyze customer profiles and financial behavior, achieving improved accuracy in loan risk prediction. This approach ensures that models do not rely solely on numerical data, reducing bias and improving generalization. Similarly, Reddy et al. (2023) confirmed that Bidirectional Long Short-Term Memory (BiLSTM) models, when combined with feature extraction methods, yield better prediction accuracy than standalone models. In this hybrid approach, deep neural networks extract key features from applicant data, while BiLSTM analyzes sequential financial patterns. Attention mechanisms highlight important factors like credit score, income stability, and debt ratio, improving accuracy and reducing misclassification.

To improve loan prediction performance, researchers have increasingly explored deep learning-based architectures. Mehta et al. (2022) enhanced deep learning-based loan prediction by incorporating advanced architectures such as TabNet and deep ensemble networks. Their study showed a significant increase in prediction accuracy, but the absence of temporal modeling limited their effectiveness in analyzing time-series financial data. Singh et al. (2021) demonstrated the effectiveness of deep neural networks in extracting complex patterns from financial data. Their study compared different architectures, including Multi-Layer Perceptron (MLP) and Deep Feedforward Networks, for predicting loan approvals. While deep learning models improved accuracy over traditional machine learning techniques, they struggled with interpretability, making it difficult for financial institutions to justify decisions. To address the

lack of temporal understanding in static models, researchers explored Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks for loan prediction. Verma et al. (2021) trained an LSTM network on customer transaction histories, demonstrating that LSTMs could effectively capture sequential dependencies in financial behavior. However, the model faced challenges with long sequences due to vanishing gradient problems, leading to reduced performance in large-scale datasets.

Before the rise of advanced deep learning methods, loan decision prediction relied heavily on machine learning models and traditional statistical approaches. Patel et al. (2020) introduced a Random Forest-based classifier for loan risk prediction, leveraging feature importance ranking to improve classification accuracy. Their approach outperformed traditional Logistic Regression models by reducing false positives in loan approval decisions. However, it lacked the ability to capture sequential financial behavior over time, such as transaction trends. Early research in loan decision prediction relied on statistical models such as Logistic Regression and Linear Discriminant Analysis (LDA). These models were widely used due to their simplicity and interpretability. Lessmann et al. (2019) conducted a comparative study of various credit scoring techniques and found that traditional statistical models provided baseline performance but lacked the ability to capture complex nonlinear relationships in financial data.

To further enhance prediction accuracy, researchers have explored the use of multi-source data, including transaction history, credit bureau data, and behavioral analytics. While most studies initially focused on static applicant data, incorporating dynamic financial behavior such as spending patterns and repayment history has improved model reliability. Studies have also investigated the use of alternative data sources, such as mobile usage and digital payment history, to assess creditworthiness, especially for individuals with limited credit history. Another promising area is transfer learning, which utilizes pre-trained models to improve prediction performance when limited data is available. Models trained on large financial datasets can be fine-tuned for specific loan prediction tasks, reducing computational costs while maintaining high accuracy. Transfer learning has proven particularly useful in scenarios where labeled financial data is scarce. Federated learning has also gained attention as a privacy-preserving technique in financial applications. Instead of sharing sensitive customer data, models are trained locally on institutional data, and only model updates are shared. This

approach ensures data privacy while maintaining high prediction performance, making it suitable for banking systems where confidentiality is critical.

Additionally, recent research has focused on Explainable AI (XAI) techniques to enhance transparency in loan decision systems. Methods such as SHAP and LIME provide insights into model predictions, enabling financial institutions to understand the factors influencing decisions. Explainability is essential in loan approval systems to ensure fairness and compliance with regulatory standards. Studies have shown that integrating XAI with machine learning models improves trust and accountability in automated decision-making systems.

Another important advancement is the use of synthetic data generation techniques to overcome data scarcity and privacy concerns. Researchers have used Generative Adversarial Networks (GANs) and data augmentation methods to generate realistic financial data for training models. These techniques improve model robustness by exposing algorithms to diverse scenarios, enhancing their performance on real-world datasets.

Most prior research has focused on predicting loan defaults based on historical financial data. However, recent studies emphasize real-time loan decision systems that consider dynamic financial behavior and external factors. Ethical concerns, data privacy, and fairness have become critical challenges in developing reliable loan prediction systems. Researchers are increasingly focusing on bias mitigation techniques and responsible AI practices to ensure fair decision-making.

Despite significant advancements, several challenges remain in loan decision systems. Traditional machine learning models lack the ability to capture sequential financial behavior. Deep learning models often suffer from high computational complexity and lack interpretability. The absence of ensemble techniques in some models limits prediction stability. To address these limitations, the proposed system integrates Random Forest, Decision Tree, Logistic Regression, and Explainable AI techniques. This hybrid approach enhances feature analysis, improves prediction accuracy, and ensures transparency in decision-making.

The literature survey highlights the evolution of loan prediction systems from traditional statistical models to advanced machine learning and deep learning approaches. By leveraging ensemble learning, attention mechanisms, and explainable AI, researchers have significantly improved prediction accuracy and reliability. The proposed system builds upon

these advancements by combining multiple machine learning models with explainable techniques to provide accurate, transparent, and efficient loan decision support. This study contributes to the development of intelligent financial systems that enhance user trust and support fair and data-driven decision-making.

The literature survey emphasizes the need for advanced machine learning models in loan decision systems. Traditional approaches struggle with accuracy and interpretability, while recent models such as ensemble learning and explainable AI provide improved performance. Although these methods show promising results, future research should focus on real-time prediction, fairness, scalability, and integration into practical banking systems.

2.1 REVIEW OF RELATED WORK

The prediction of loan approval and credit risk has been widely studied in financial analytics and machine learning. Various approaches have been proposed, ranging from traditional statistical methods to advanced AI-based systems. This section reviews the major techniques used in loan prediction systems.

1. Traditional Loan Decision Approaches

Early loan approval systems relied on manual evaluation and rule-based decision-making. Bank officers assessed applicants based on predefined criteria such as income, employment, and credit score. While these systems were simple and interpretable, they were time-consuming and prone to human bias. Rule-based systems lacked flexibility and could not adapt to changing financial patterns. In many cases, the final decision depended heavily on the experience and judgment of the loan officer. These approaches were also inefficient when handling a large number of applications. Since they followed fixed rules, they often failed to identify hidden risk patterns in customer data. As a result, traditional methods were less accurate and less suitable for modern digital banking environments.

2. Machine Learning-Based Approaches

Machine learning models have significantly improved loan prediction accuracy. Algorithms such as Logistic Regression, Decision Trees, and Support Vector Machines have been widely used for credit risk assessment. These models analyze historical data to identify patterns associated with loan repayment behavior. However, they rely heavily on feature engineering and may not perform well with complex datasets. Random Forest, an ensemble learning technique, has been widely adopted due to its robustness and ability to handle high-dimensional data. Studies have shown that Random Forest outperforms traditional classifiers by reducing overfitting and improving generalization.

3. Deep Learning-Based Approaches

Deep learning models, particularly Artificial Neural Networks (ANNs), have been used to capture complex nonlinear relationships in financial data. These models provide higher accuracy compared to traditional methods but require large datasets and computational resources. Deep learning models also lack interpretability, making them less suitable for financial applications where transparency is required.

4. Explainable AI and Fairness

Explainability is a critical requirement in loan decision systems. Techniques such as LIME and SHAP have been widely used to provide explanations for model predictions. These methods help both banks and applicants understand the factors influencing loan decisions, improving trust and transparency. They also support financial institutions in meeting regulatory and compliance requirements. By identifying the most influential features, explainable models help reduce bias in decision-making. This improves fairness and accountability in automated loan approval systems. As a result, explainability plays an important role in making AI-based financial systems more reliable and acceptable.

Recent research has also focused on fairness in AI models. Bias detection and mitigation techniques are used to ensure that loan decisions are not influenced by sensitive attributes such as gender or ethnicity.

5. Comparison with the Proposed Approach

While ehile existing systems provide good accuracy, they often lack transparency and fairness. The proposed system integrates multiple machine learning models with explainable AI techniques to provide accurate, transparent, and unbiased loan decisions. This approach ensures better performance compared to traditional methods.

2.2 DEFINITION OF PROBLEM STATEMENT

The primary objective of this project is to develop an intelligent and automated loan decision system that can accurately predict whether a loan application should be approved or rejected. The system aims to analyze applicant data such as income, credit history, employment status, and financial behavior to determine creditworthiness. The key challenges include handling large datasets, ensuring model accuracy, reducing bias, providing explainable decisions, and enabling real-time predictions. The system also aims to enhance transparency and fairness in loan approval processes, ensuring that decisions are understandable and justifiable..

2.3 EXISTING SYSTEM

The Existing loan decision systems use a combination of statistical methods and machine learning algorithms. Traditional systems rely on Logistic Regression, Decision Trees, and Support Vector Machines to classify loan applications. These systems use features such as income, credit score, and employment history to make predictions.

Some systems incorporate ensemble learning techniques such as Random Forest and Gradient Boosting to enhance prediction accuracy and overall model performance. By combining multiple algorithms, these methods reduce errors and improve the reliability of predictions compared to single models. In addition to performance improvements, recent systems also integrate explainable AI techniques to provide clear insights into the decision-making process, helping both applicants and institutions understand the reasons behind approvals or rejections. According to the AI-Lend Sense project, loan prediction models commonly utilize algorithms such as Logistic Regression, Decision Tree, and Random Forest to analyze applicant data, including income, credit history, and financial behavior, and generate accurate and transparent loan approval decisions.

Limitations of Existing System

Despite significant advancements in loan prediction and credit risk assessment, existing systems still suffer from several challenges:

- **Inability to capture complex financial relationships:** Traditional machine learning models such as Logistic Regression and Decision Trees often assume linear or simple relationships between variables. However, real-world financial data is highly complex and nonlinear. Studies by Ahmed et al. show that these models fail to capture hidden patterns in customer behavior, leading to inaccurate predictions in certain cases.

- **High false approval and rejection rates**: Many existing systems struggle to balance precision and recall, resulting in incorrect decisions. For example, a creditworthy applicant may be rejected (false negative), or a risky applicant may be approved (false positive). According to Mehta and Sharma , this issue arises due to imbalanced datasets and insufficient feature representation, affecting the reliability of predictions. Such misclassifications can lead to financial losses for banks and reduce trust among genuine applicants. A high false approval rate increases the chances of loan defaults and non-performing assets (NPAs), while a high false rejection rate may cause institutions to lose potentially reliable customers and business opportunities. Therefore, improving the model's ability to accurately distinguish between eligible and risky applicants is essential for ensuring a fair, efficient, and reliable loan approval system.

- **Computational inefficiency**: Training complex models on large-scale financial datasets requires significant computational resources. Deep learning models, in particular, demand high processing power and memory, making real-time loan approval challenging.
- **Overfitting and poor generalization:** Many models perform well on training datasets but fail to generalize to new or unseen data. This leads to decreased performance in real-world scenarios.

2.4 PROPOSED SYSTEM

The proposed system introduces an AI-based loan decision framework that integrates multiple machine learning algorithms, including Logistic Regression, Decision Tree, and Random Forest, along with Explainable AI techniques such as LIME. The system analyzes applicant data, including income, credit score, employment status, and financial history, to predict loan approval. The use of ensemble methods improves accuracy and reduces overfitting. Explainable AI techniques provide clear insights into the factors influencing each decision, ensuring transparency and trust. The system also includes a user-friendly web interface that allows applicants to input their data and receive instant predictions along with explanations. This improves user experience and helps applicants understand their eligibility.

Advantages of the Proposed System:

The proposed loan decision system significantly improves upon existing approaches by addressing key limitations in traditional and machine learning-based credit risk prediction systems:

- **Higher Accuracy**: Combining Logistic Regression, Decision Tree, and Random Forest with Explainable AI improves prediction accuracy by reducing false positives and negatives.
- **Improved Feature Understanding:** Using multiple financial factors like income, credit history, and debt ratio helps capture hidden patterns and reduces misclassification.
- **Better Generalization & Robustness**: Ensemble methods like Random Forest reduce overfitting and perform consistently on new data.
- **Efficient Processing**: ML models such as Decision Tree and Random Forest provide faster training and real-time predictions.
- **Enhanced Interpretability**: Explainable AI techniques like LIME improve transparency by explaining loan decisions.

2.5 OBJECTIVES

- **Develop an Automated Loan Detection System** – Develop an ML model to automatically approve or reject loan applications based on applicant data.
- **Efficient Decision - Making**– Reduce manual work and ensure faster, consistent loan processing using AI techniques.
- **Improve Accuracy** – Use ensemble models like Random Forest, Decision Tree, and Logistic Regression to increase prediction accuracy.
- **Ensure Scalability** – Design the system to handle large volumes of applications with real-time processing.
- **Transparency & Fairness** – Apply Explainable AI to provide clear, unbiased, and trustworthy loan decisions.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements,

- Processor : Intel Core i3 or higher.
- Hard Disk : 20GB or above.
- RAM : 4GB or above.

2.6.2 SOFTWARE REQUIREMENTS:

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are some software requirements,

- Operating system : Windows 10
- Language : Python
- Tools : Jupyter Notebook/ VS Code/ Notepad
- Frame Work : Flask / Django
- Libraries : Scikit – Learn, Pandas, Numpy.

3. SYSTEM ARCHITECTURE & DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

Project architecture refers to the structural framework and design of a system, defining its components, interactions, and overall workflow. It provides a clear blueprint for development, ensuring scalability, efficiency, and transparency. An effective architecture supports the entire lifecycle of the project, from data collection to final decision-making, reducing complexity and improving system reliability.

3.1 PROJECT ARCHITECTURE

The architecture of AI-Lend Sense shows the workflow from user input to loan prediction and explanation, combining machine learning and explainable AI to ensure accurate, fair, and transparent decisions.

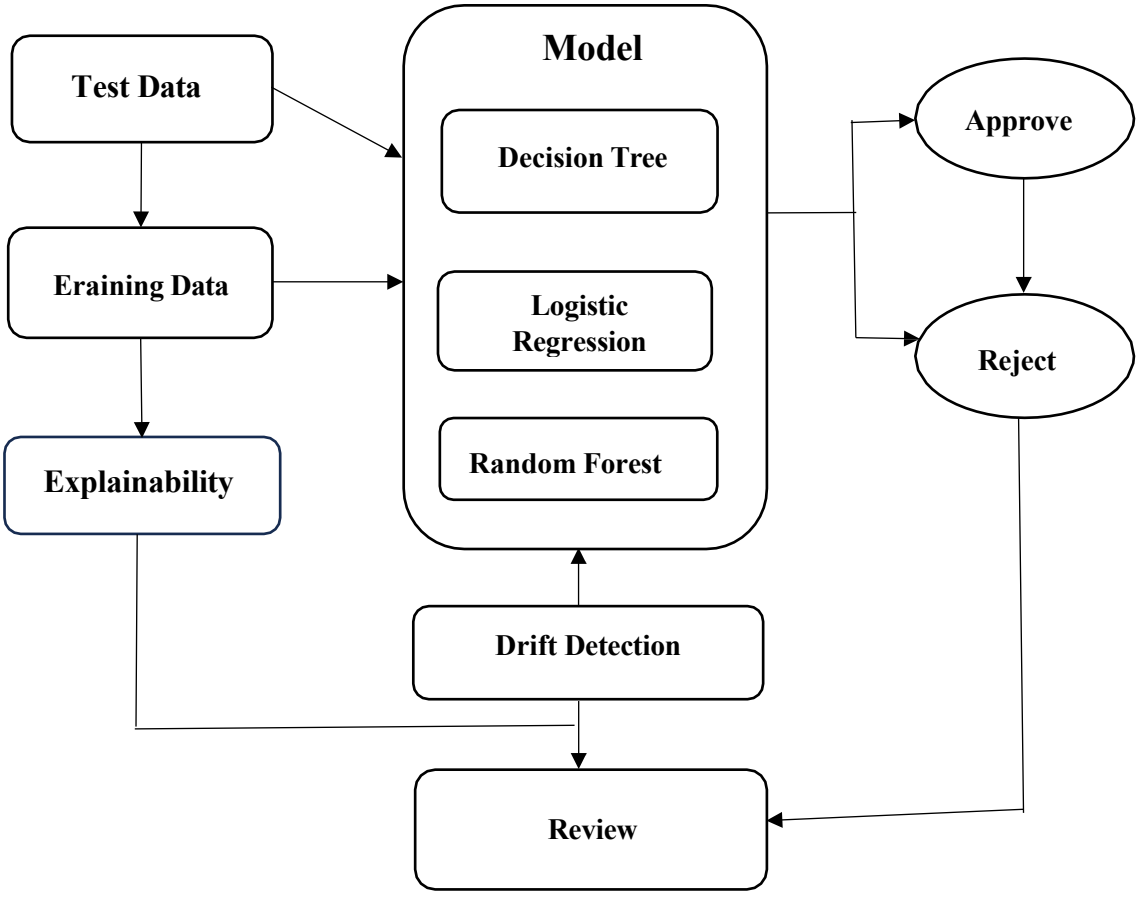

Figure 3.1: Project Architecture of Ai – Lend Sense: Predictive And Explainable Loan Decision System.

3.2 DESCRIPTION

Input Data: Basic applicant details like income, job, credit score, loan amount, and financial history are used to assess loan eligibility.

Reading Data: Data is collected from sources (datasets or bank records), cleaned, and prepared using Python libraries for analysis.

Feature Processing: Important features (income, debt ratio, credit history) are selected and normalized to improve model accuracy.

Model Building: Machine learning models like Logistic Regression, Decision Tree, and Random Forest are used to predict loan approval.

Explainable AI: LIME explains the model's decisions by showing which factors influenced approval or rejection.

Classification Layer: The system classifies applications as Approved or Rejected using trained models.

Training and Evaluation: Models are trained on past data and evaluated using accuracy; Random Forest often performs best.

Feedback: User and admin feedback helps improve model accuracy and provides suggestions to increase approval chances.

3.3 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) represents how data moves within the system, showing processes, data stores, and external entities. It helps in understanding the flow of information and improves system design and efficiency.

A Data Flow Diagram comprises Four primary elements:

- External Entities: Applicants and bank administrators who provide and receive data.
- Processes: Data preprocessing, model training, prediction, and explanation generation.
- Data Flows: Movement of applicant data through different system modules.
- Data Stores: Databases storing loan datasets, user data, and model outputs.

These elements are represented using standard symbols such as rectangles for entities, circles for processes, arrows for data flow, and open rectangles for data storage.

Benefits:

DFDs provide a clear visualization of system functionality, making it easier for both technical and non-technical users to understand the workflow. They help identify inefficiencies, improve data handling, and ensure transparency in the loan decision process.

Applications:

DFDs are widely used in financial systems, banking applications, and AI-based decision systems. They help in analyzing workflows, improving system performance, and ensuring secure and efficient data processing.

Levels of DFD:

DFDs are divided into different levels for better understanding:

- Level 0 (Context Diagram): Shows the overall system with input and output interactions.
- Level 1: Breaks down the system into major processes like data preprocessing, model training, and prediction.
- Level 2+: Provides detailed representation of individual modules for complex systems.

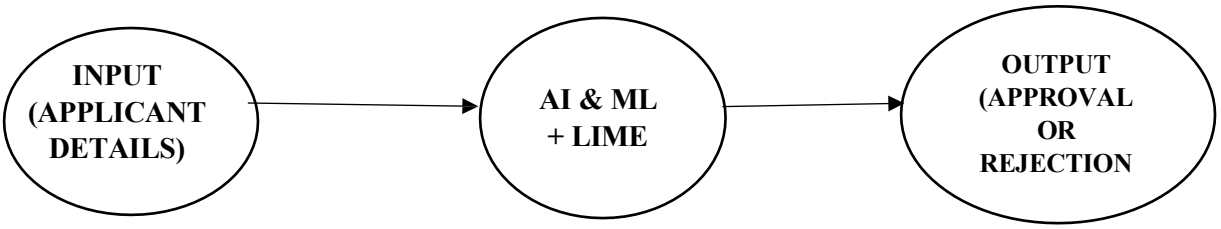

Figure 3.2: Dataflow Diagram of Ai – Lend Sense: Predictive And Explainable Loan Decision System.

4. IMPLEMENTATION

4. IMPLEMENTATION

Now-a-days banks and financial institutions are handling large number of loan applications from different types of customers. This system provides financial services such as loan approval based on customer details like income, employment, credit score and transaction history. However, manual evaluation of loan applications is time consuming and may lead to errors and unfair decisions. Some applicants may be rejected or approved without proper analysis which affects both banks and customers.

Such incorrect decisions may cause financial loss to banks and reduce trust among customers, so there is a need for an intelligent system to predict loan decisions accurately before approval. To prevent this issue, we proposed a system by employing Machine Learning algorithms to predict loan approval and credit risk. Logistic Regression, Decision Tree and Random Forest algorithms are used to analyze the applicant data and predict whether the loan will be approved or rejected.

In proposed work, a novel AI based architecture is developed for loan prediction and classification. The system uses historical loan data including applicant income, credit history, employment details and loan amount to train the model. These features are processed and fed into multiple machine learning algorithms to learn patterns and improve prediction accuracy. Explainable AI technique such as LIME is also integrated to provide clear explanation for each prediction, showing the factors affecting loan approval or rejection.

Multiple algorithms such as Logistic Regression, Decision Tree and Random Forest are trained on the loan dataset and evaluated. Among these, Random Forest algorithm gives better accuracy compared to other models. We also compared the results with traditional methods, but machine learning based models provide better performance and faster decision making.

In this project, real world loan datasets are used instead of sensitive or restricted data. The dataset is divided into categories such as approved and rejected loans. Based on prediction results, the system classifies applications into safe (eligible) and risky (non-eligible) categories. This helps banks to reduce financial risk and ensures fair and transparent loan decision process for customers.

In addition to improving prediction accuracy, the proposed loan decision system also supports better operational efficiency in banking environments. By automating the evaluation process, the system reduces the workload of bank staff and minimizes the time required to process each application. This allows financial institutions to handle a larger number of loan requests in a shorter period, making the overall process faster and more scalable. The automated system also ensures consistency in decision-making by applying the same evaluation criteria to all applicants, thereby reducing the possibility of human inconsistency.

Another important advantage of this system is its ability to support data-driven decision-making. Instead of relying only on manual judgment, the model uses patterns learned from historical loan records to estimate the probability of loan approval. This enables banks to make more informed decisions based on actual applicant behavior and financial indicators. The system can also help identify high-risk applications at an early stage, which supports better credit risk management and reduces the chances of loan default.

The proposed model is designed in a way that it can be easily integrated into modern digital banking platforms and financial applications. It can be used as a decision support tool for loan officers or as part of an online loan application portal where customers submit their details digitally. Once the user enters the required information, the model can instantly predict the likely loan outcome and provide supporting reasons for that prediction. This improves user experience by making the loan process more transparent and responsive.

Data preprocessing plays an important role in the performance of the system. Before training the models, missing values, inconsistent entries, and categorical variables are handled carefully to improve data quality. Techniques such as encoding, normalization, and feature selection are applied so that the machine learning algorithms can work effectively on the dataset. Proper preprocessing helps improve model stability and ensures that the system performs well even when real-world data contains noise or incomplete records. Overall, the proposed loan prediction and classification system provides an efficient, transparent, and reliable solution for modern financial institutions. It not only helps improve loan approval decisions but also supports risk reduction, fairness, and faster service delivery. Therefore, this system can play an important role in transforming traditional loan processing into a more intelligent and automated decision-making framework.

We used a loan dataset with applicant details like income, credit score, loan amount, and employment status, classified as Loan Approved or Loan Rejected.

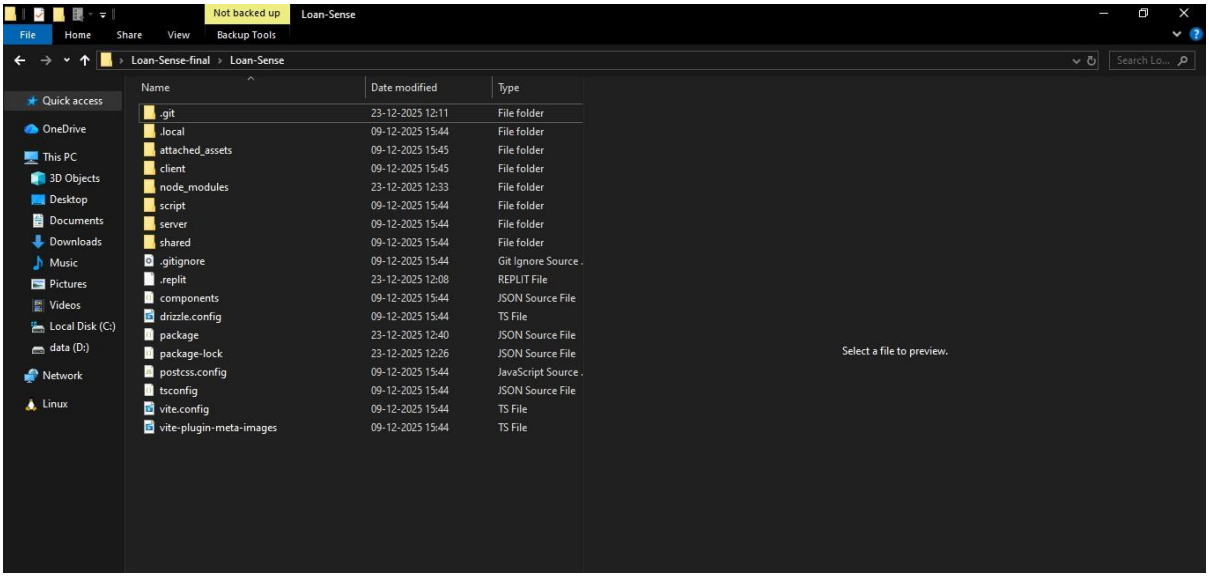

Figure 4.1: Dataset directory structure showing the Loan-Sense Project Root Folder containing client, server, shared modules and configuration files.

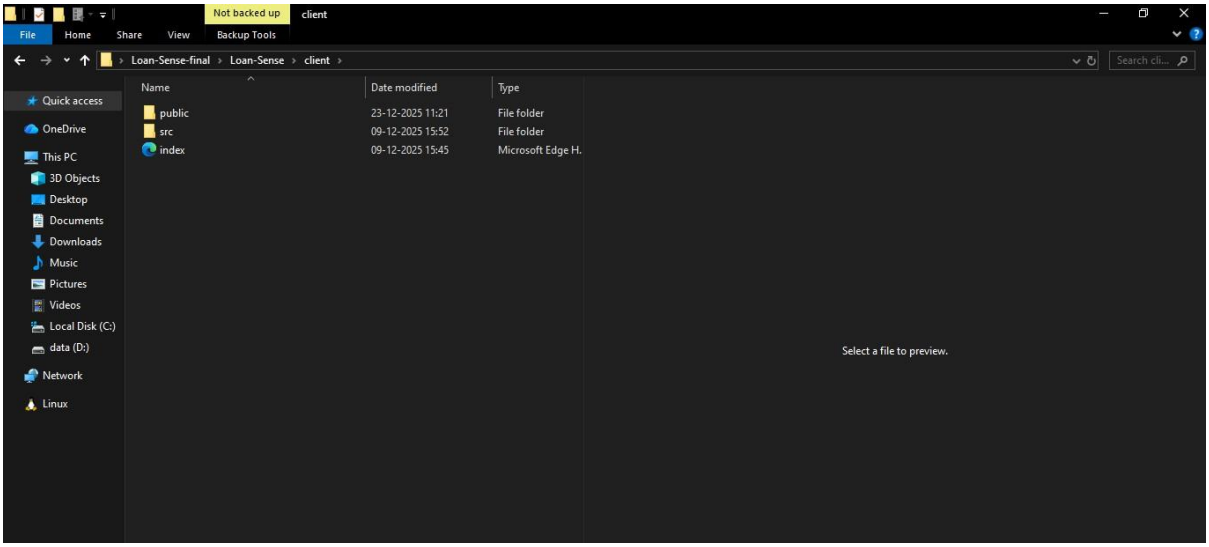

Figure 4.2: Screenshot of the "Client" Folder Showing Frontend Application Files and Components.

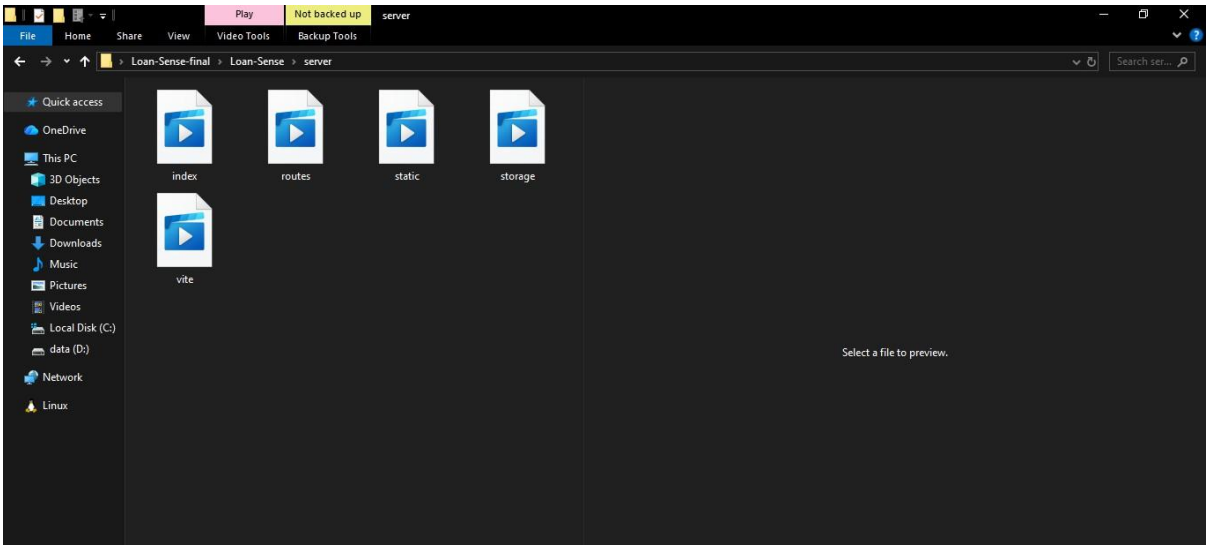

Figure 4.3: Screenshot of the "Server" Folder Showing Backend API, Routes and Database Configuration Files.

To implement this project we have designed following modules:

1) **Upload Loan Application Dataset:** This module allows uploading a loan dataset containing applicant details such as income, employment status, credit history, and loan amount. The dataset is preloaded in the system, and during execution, users can manually enter or select input values from it**.**
2) **Dataset Preprocessing:** Read data, handle missing values, encode categorical features, normalize data, and shuffle dataset for better training.
3) **Run Proposed ML–Explainable AI Algorithm**: Split data into train/test, train models (Logistic Regression, Decision Tree, Random Forest), use LIME for explanations, and evaluate accuracy.
4) **Run Logistic Regression Algorithm**: Train Logistic Regression model and calculate prediction accuracy on test data.
5) **Run Random Forest Algorithm:** Train Random Forest model and evaluate prediction accuracy.
6) **Comparison Graph:** Plot accuracy comparison between proposed model and existing models.

4.1 SAMPLE CODE

DATASET.TS

```
export interface KaggleLoanData {
  Loan_ID: string;
  Gender: "Male" | "Female";
  Married: "Yes" | "No";
  Dependents: "0" | "1" | "2" | "3+";
  Education: "Graduate" | "Not Graduate";
  Self_Employed: "Yes" | "No";
  ApplicantIncome: number;
  CoapplicantIncome: number;
  LoanAmount: number;
  Loan_Amount_Term: number;
  Credit_History: 1 | 0;
  Property_Area: "Urban" | "Semiurban" | "Rural";
  Loan_Status: "Y" | "N";
}
export const DATASET_COLUMNS = [
  "Loan_ID",
  "Gender",
  "Married",
  "Dependents",
  "Education",
  "Self_Employed",
  "ApplicantIncome",
  "CoapplicantIncome",
  "LoanAmount",
  "Loan_Amount_Term",
  "Credit_History",
  "Property_Area",
  "Loan_Status"
```

```
];

export const LOAN_DATASET: KaggleLoanData[] = [
  {
    Loan_ID: "LP001002",
    Gender: "Male",
    Married: "No",
    Dependents: "0",
    Education: "Graduate",
    Self_Employed: "No",
    ApplicantIncome: 5849,
    CoapplicantIncome: 0,
    LoanAmount: 140,
    Loan_Amount_Term: 360,
    Credit_History: 1,
    Property_Area: "Urban",
    Loan_Status: "Y"
  }
];
```

CLIENT SIDE

```
import { KaggleLoanData } from
"./dataset"; export interface
ApplicantData {
fullName:
string;
email:
string;
phone:
string; pan:
string;
aadhar:
string;
bankAccount:
string; ifsc:
```

```
loanAmount: number;
loanTerm: number;
debtToIncome:
number; purpose:
string;
}
export interface
PredictionResult { id: string;
timestamp: string;
applicant:
ApplicantData;
decision: 'Approved' | 'Rejected';
probability: number;
riskLevel: 'Low' | 'Medium' | 'High';
}
        modelConsensus: {
        logisticRegression: {
                probability: number;
                vote: 'Approved' | 'Rejected';
        };
        randomForest: {
        probability: number;
                vote: 'Approved' | 'Rejected';
        };
        decisionTree: {
        probability: number;
                vote: 'Approved' | 'Rejected';
        };
        };

        limeExplanatio
        n: { feature:
        string; weight:
        number; value:
```

```
fairnessMetrics: {
   demographicParity: number;
   equalOpportunity: number;
   biasDetected: boolean;
};

actionableFeedback: string[];
}
export const predictLoanEligibility = (
   data: ApplicantData
): Promise<PredictionResult> => {
   return new Promise((resolve) => {
      setTimeout(() => {
         const normIncome = Math.min(data.income / 100000, 1);
         const normCredit = (data.creditScore - 300) / 550;
         const normDTI = 1 - Math.min(data.debtToIncome, 1);
         const normLoanAmt = 1 - Math.min(data.loanAmount / 50000, 1);
         const lrScore =
            (normCredit * 0.5) +
            (normIncome * 0.3) +
            (normDTI * 0.2);

         const lrProb = Math.min(
            Math.max(lrScore + (Math.random() * 0.1 - 0.05), 0),
            1
         );
         let dtProb = 0.5;
         if (data.creditScore < 600) {
            dtProb = 0.1;
         } else if (data.debtToIncome > 0.45) {
            dtProb = 0.2;
         } else if (data.income > 50000) {
            dtProb = 0.9;
```

```
} else {
  dtProb = 0.7;
}
const rfScore =
  (normCredit * 0.4) +
  (normDTI * 0.3) +
  (normLoanAmt * 0.2) +
  ((data.employmentYears / 20) * 0.1);

const rfProb = Math.min(
  Math.max(rfScore + (Math.random() * 0.05 - 0.025), 0),
  1
);
const finalProbability =
  (lrProb * 0.3) +
  (dtProb * 0.2) +
  (rfProb * 0.5);

const decision = finalProbability > 0.6 ? 'Approved' : 'Rejected';
const features = [
  {
    feature: 'Credit History',
    weight: ((data.creditScore - 650) / 100) * 0.35,
    value: `${data.creditScore}`,
    impactDescription:
      data.creditScore > 700
        ? "Strong credit history significantly boosts approval odds."
        : "Credit score is below preferred threshold."
  },
  {
    feature: 'Debt-to-Income',
    weight: (0.35 - data.debtToIncome) * 0.4,
    value: `${(data.debtToIncome * 100).toFixed(0)}%`,
    impactDescription:
```

```
        data.debtToIncome < 0.3
          ? "Low debt ratio shows good financial health."
          : "High debt obligation relative to income is a risk factor."
    },
    {
      feature: 'Loan Amount',
      weight: ((data.income * 0.4 - data.loanAmount) / 10000) * 0.1,
      value: `₹${data.loanAmount.toLocaleString()}`,
      impactDescription:
        data.loanAmount > data.income * 0.5
          ? "Loan amount is high compared to annual income."
          : "Loan amount is well within affordable limits."
    },
    {
      feature: 'Employment Stability',
      weight: (data.employmentYears - 2) * 0.05,
      value: `${data.employmentYears} years`,
      impactDescription:
        data.employmentYears > 5
          ? "Long-term employment indicates stability."
          : "Short employment history adds uncertainty."
    }
  ];

  const limeExplanation = features.sort(
    (a, b) => Math.abs(b.weight) - Math.abs(a.weight)
  );
  const actionableFeedback: string[] = [];
  if (data.creditScore < 700) {
    actionableFeedback.push("Improve credit score by 30+ points.");
  }
  if (data.debtToIncome > 0.35) {
    actionableFeedback.push("Reduce DTI below 30%.");
  }
```

```
if (data.loanAmount > data.income * 0.4) {
    actionableFeedback.push("Reduce requested loan amount.");
}
if (actionableFeedback.length === 0) {
    actionableFeedback.push("Profile is strong; maintain current status.");
}

resolve({
    id: Math.random().toString(36).substring(7).toUpperCase(),
    timestamp: new Date().toISOString(),
    applicant: data,
    decision,
    probability: finalProbability,
    riskLevel:
        finalProbability > 0.8
            ? 'Low'
            : finalProbability > 0.6
            ? 'Medium'
            : 'High',
    modelConsensus: {
        logisticRegression: {
            probability: lrProb,
            vote: lrProb > 0.6 ? 'Approved' : 'Rejected'
        },
        randomForest: {
            probability: rfProb,
            vote: rfProb > 0.6 ? 'Approved' : 'Rejected'
        },
        decisionTree: {
            probability: dtProb,
            vote: dtProb > 0.6 ? 'Approved' : 'Rejected'
        }
    },
```

```
            limeExplanation,
            fairnessMetrics: {
                demographicParity: 0.98,
                equalOpportunity: 0.95,
                biasDetected: false
            },
            actionableFeedback
        });
    }, 2500);
  });
};
```

SERVER SIDE

```
import express, { type Request, type Response, type NextFunction } from "express";
import { createServer, type Server } from "http";
import type { Express } from "express";
import { randomUUID } from "crypto";

import { registerRoutes } from "./routes";
import { serveStatic } from "./static";
import { storage } from "./storage";
import { type User, type InsertUser } from "@shared/schema";
const app = express();
const httpServer = createServer(app);
declare module "http" {
  interface IncomingMessage {
    rawBody: unknown;
  }
}
app.use(
  express.json({
    verify: (req, _res, buf) => {
```

```
            req.rawBody = buf;
        }
    })
);

app.use(express.urlencoded({ extended: false }));
export function log(message: string, source = "express") {
    const t = new Date().toLocaleTimeString("en-US", {
        hour: "numeric",
        minute: "2-digit",
        second: "2-digit",
        hour12: true
    });
    console.log(`${t} [${source}] ${message}`);
}

app.use((req, res, next) => {
    const start = Date.now();
    const path = req.path;
    let captured: Record<string, any> | undefined;
    const original = res.json;
    res.json = function (body, ...args) {
        captured = body;
        return original.apply(res, [body, ...args]);
    };
    res.on("finish", () => {
        const duration = Date.now() - start;
        if (path.startsWith("/api")) {
            let line = `${req.method} ${path} ${res.statusCode} in ${duration}ms`;
            if (captured) {
                line += ` :: ${JSON.stringify(captured)}`;
            }
            log(line);
        }
```

```
    });

    next();
});
(async () => {
    await registerRoutes(httpServer, app);
    app.use((err: any, _req: Request, res: Response, _next: NextFunction) => {
        const status = err.status || err.statusCode || 500;
        const message = err.message || "Internal Server Error";
        res.status(status).json({ message });
    });
    if (process.env.NODE_ENV === "production") {
        serveStatic(app);
    } else {
        const { setupVite } = await import("./vite");
        await setupVite(httpServer, app);
    }
    const port = Number(process.env.PORT) || 3000;
    httpServer.listen(port, () => {
        log(`Server running on http://localhost:${port}`);
    });
})();
export async function registerRoutes(
    httpServer: Server,
    app: Express
): Promise<Server> {
    return httpServer;
}
export interface IStorage {
    getUser(id: string): Promise<User | undefined>;
    getUserByUsername(username: string): Promise<User | undefined>;
    createUser(user: InsertUser): Promise<User>;
}
export class MemStorage implements IStorage {
```

```
  private users: Map<string, User>;

  constructor() {
    this.users = new Map();
  }
  async getUser(id: string): Promise<User | undefined> {
    return this.users.get(id);
  }
  async getUserByUsername(username: string): Promise<User | undefined> {
    return Array.from(this.users.values()).find(
      (u) => u.username === username
    );
  }
  async createUser(insertUser: InsertUser): Promise<User> {
    const id = randomUUID();
    const user: User = {
      ...insertUser,
      id
    };

    this.users.set(id, user);
    return user;
  }
}

export const storage = new MemStorage();
```

VITE PLUGIN

```
import type { Plugin } from "vite";
import fs from "fs";
import path from "path";
export function metaImagesPlugin(): Plugin {
  return {
```

```
name: "vite-plugin-meta-images",

transformIndexHtml(html) {
  const baseUrl = getDeploymentUrl();
  if (!baseUrl) {
    log("[meta-images] no domain found");
    return html;
  }
  const publicDir = path.resolve(process.cwd(), "client", "public");
  const png = path.join(publicDir, "opengraph.png");
  const jpg = path.join(publicDir, "opengraph.jpg");
  const jpeg = path.join(publicDir, "opengraph.jpeg");
  let ext: string | null = null;
  if (fs.existsSync(png)) {
    ext = "png";
  } else if (fs.existsSync(jpg)) {
    ext = "jpg";
  } else if (fs.existsSync(jpeg)) {
    ext = "jpeg";
  }

  if (!ext) {
    log("[meta-images] image not found");
    return html;
  }
  const imageUrl = `${baseUrl}/opengraph.${ext}`;
  log("[meta-images] updating:", imageUrl);
  html = html.replace(
    /<meta\s+property="og:image"\s+content="[^"]*"\s*\/>/g,
    `<meta property="og:image" content="${imageUrl}" />`
/<meta\s+property="og:image"\s+content="[^"]*"\s*\/>/g,
    `<meta property="og:image" content="${imageUrl}" />`

  );
```

```
            html = html.replace(
              /<meta\s+name="twitter:image"\s+content="[^"]*"\s*\/>/g,
              `<meta name="twitter:image" content="${imageUrl}" />`
            );

            return html;
        }
    };
}
function getDeploymentUrl(): string | null {
    if (process.env.REPLIT_INTERNAL_APP_DOMAIN) {
        return `https://${process.env.REPLIT_INTERNAL_APP_DOMAIN}`;
getUser(id: string): Promise<User | undefined>;
    getUserByUsername(username: string): Promise<User | undefined>;
    createUser(user: InsertUser): Promise<User>;

    }

    if (process.env.REPLIT_DEV_DOMAIN) {
        return `https://${process.env.REPLIT_DEV_DOMAIN}`;
    }

    return null;
}

function log(...args: any[]): void {
    if (process.env.NODE_ENV === "production") {
        console.log(...args);
    }
}
```

VITE CONFIG

```
import { defineConfig } from "vite";
import react from "@vitejs/plugin-react";
import tailwindcss from "@tailwindcss/vite";
import path from "path";
import runtimeErrorOverlay from "@replit/vite-plugin-runtime-error-modal";
import { metaImagesPlugin } from "./vite-plugin-meta-images";

export default defineConfig({
  plugins: [
    react(),
    runtimeErrorOverlay(),
    tailwindcss(),
    metaImagesPlugin(),
    ...(process.env.NODE_ENV !== "production" &&
    process.env.REPL_ID !== undefined
      ? [
          await import("@replit/vite-plugin-cartographer").then((m) =>
            m.cartographer()
          ),
          await import("@replit/vite-plugin-dev-banner").then((m) =>
            m.devBanner()
          )
        ]
      : [])
  ],

  resolve: {
    alias: {
      "@": path.resolve(import.meta.dirname, "client", "src"),
      "@shared": path.resolve(import.meta.dirname, "shared"),
      "@assets": path.resolve(import.meta.dirname, "attached_assets")
    }
  },

  css: {
    postcss: {
      plugins: []
    }
  },

  root: path.resolve(import.meta.dirname, "client"),

  build: {
    outDir: path.resolve(import.meta.dirname, "dist/public"),
    emptyOutDir: true
  },
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

The following screenshots showcase the results of our project, highlighting key features and functionalities. These visual representations provide a clear overview of how the system performs under various conditions, demonstrating its effectiveness and user interface. The screenshots serve as a visual aid to support the project's technical and operational achievements.

5.1 Home Page :

In below screen, we can see a Home Page that describes a main Theme of Project.

When we click on Check Eligibility or else Get Started button it takes to Loan Application Form Page to upload a details.

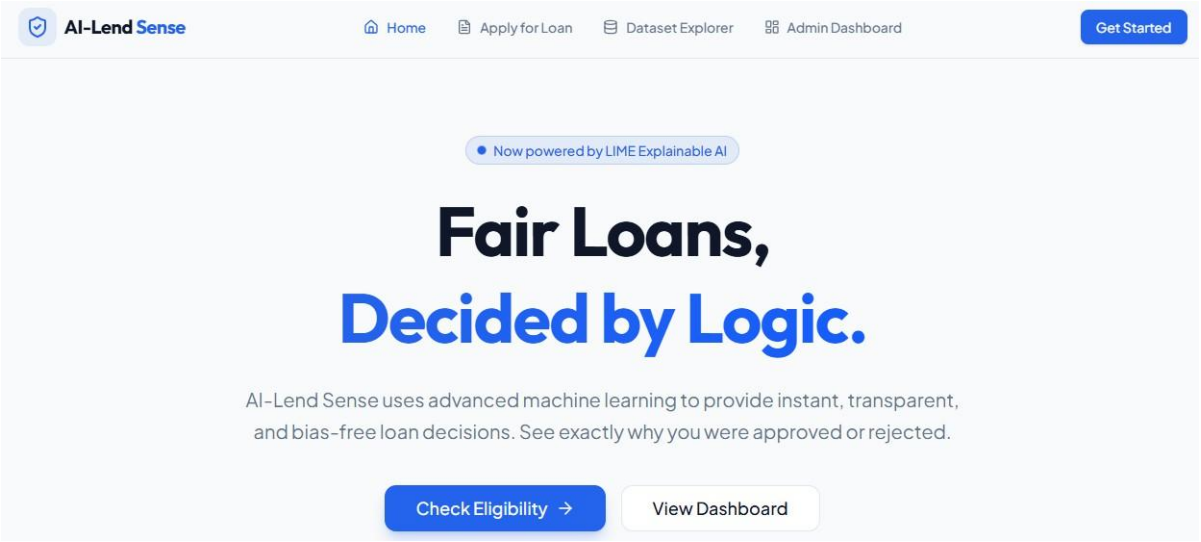

Figure 5.1 : Home Page Of AI – Lend Sense: Predictive And Explainable Loan Decision System.

5.2 Loan Application Form → Personal Information:

- This Loan Application mainly contains three Parts Personal Information, Verification, Financial Profile.

In Below Page we can see the Personal Information Of that particular person.

In this Page It asks Information like→ Full Name

→Email

→ Mobile Number etc a basic information.

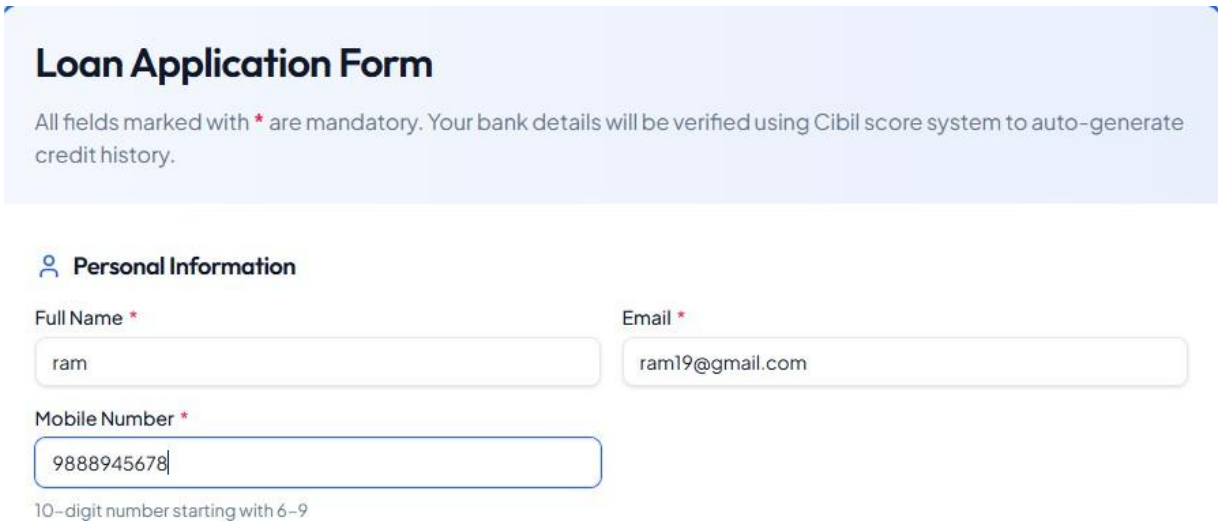

Figure 5.2 : Personal Information to upload in Loan Application Form of AI- Lend Sense: Predictive and Explainable Loan Decision System.

5.3 Loan Application Form →Verification (Govt Id & Bank Profile):

In below page, we can see the details of Government Id &.Verification and also like Bank Account details to verify to access all the details of a particular person.

In this Page It asks Information like→ Government ID : PAN CARD & AADHAR

→ Bank Details : Account Number & IFSC Code.

For each field, the system provides clear validation rules, specifying the required number of digits and the correct format to ensure that the information is entered properly.

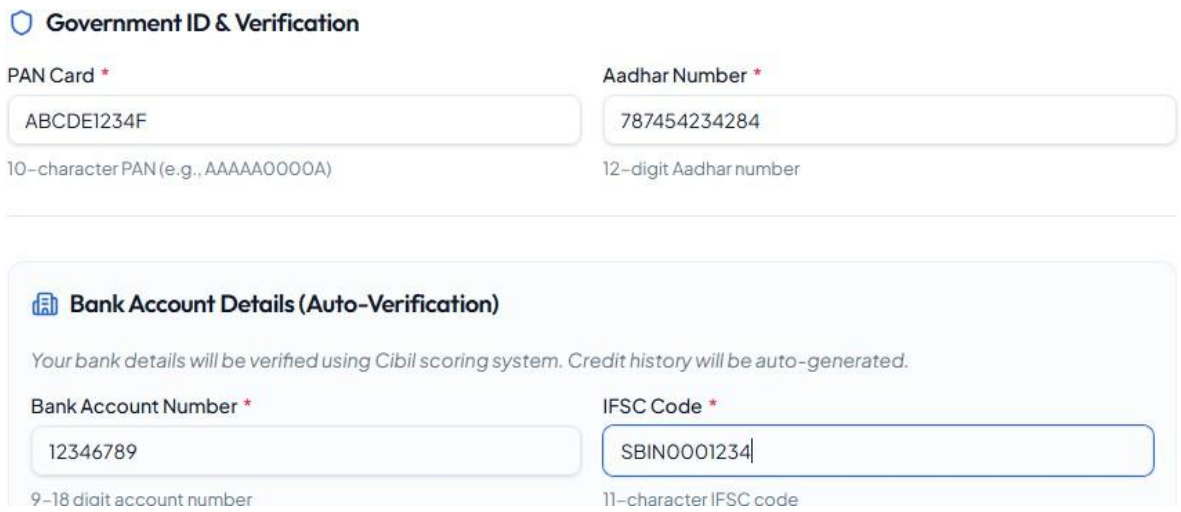

Figure 5.3 : Verification Page to upload in Loan Application Form of AI- Lend Sense: Predictive and Explainable Loan Decision System.

5.4 Loan Application Form → Financial Profile:

In below page, we can observe the **Financial Profile** page of the proposed loan prediction system **AI-Lend Sense**. This page is used to collect important financial and employment details required for loan eligibility prediction.

- **Annual Income (₹):**
  Shows the applicant's yearly income to assess repayment capacity.
- **Years Employed:**
  Indicates job stability based on total years of employment.
- **Debt-to-Income Ratio (%):**
  Shows the percentage of debt compared to monthly income.
- **Requested Loan Amount (₹):**
  Specifies the loan amount applied by the user.
- **Tenure (Months):**
  Indicates the selected repayment duration.
- **Loan Purpose:**
  Specifies the reason for applying for the loan (personal, education, home, etc.).

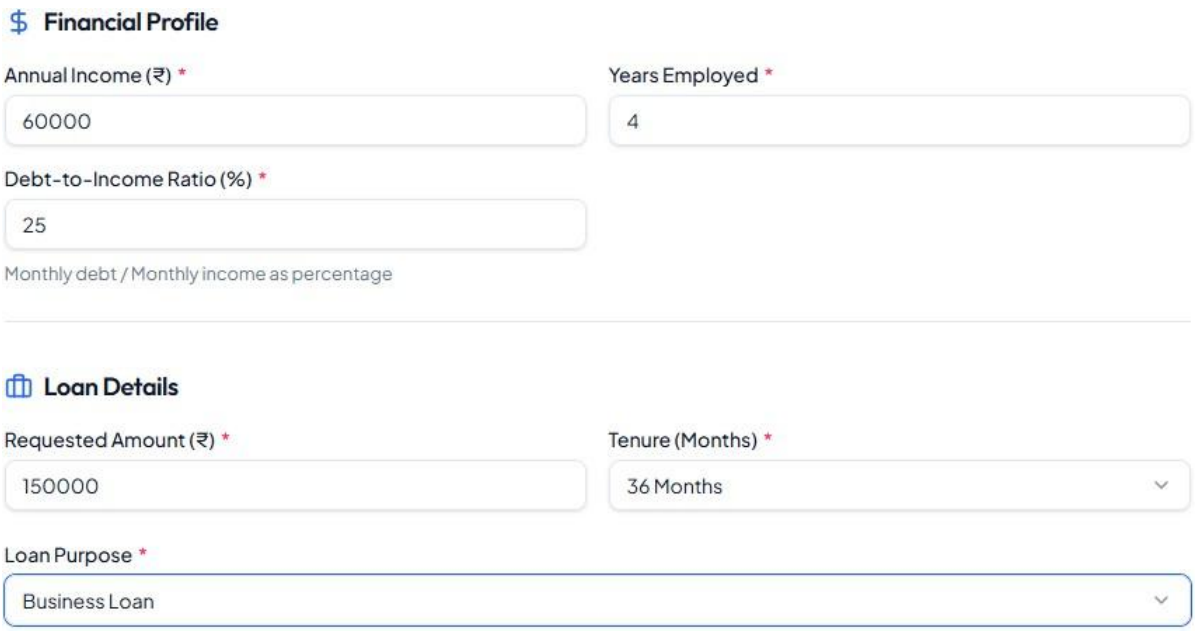

Figure 5.4 : Financial Profile Page to upload in Loan Application Form of AI- Lend Sense: Predictive and Explainable Loan Decision System.

5.5 Dataset Explorer :

In below page, It shows the Dataset Explorer page where we can either manually enter applicant details or use a preloaded dataset for training and testing the model

The dataset contains details such as Loan ID, Gender, Marital Status, Number of Dependents, Education, Self-Employment Status, Applicant Income, Co-applicant Income, Loan Amount, Loan Amount Term, Credit History, and Property Area etc.

These attributes are used by the system to analyse and predict whether the loan will be approved or rejected.

Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
LP001002	Male	No	0	Graduate	No	₹58,49,000	₹0	₹1,40,000	360m	1
LP001003	Male	Yes	1	Graduate	No	₹45,83,000	₹15,08,000	₹1,28,000	360m	1
LP001005	Male	Yes	0	Graduate	Yes	₹30,00,000	₹0	₹66,000	360m	1
LP001006	Male	Yes	0	Not Graduate	No	₹25,83,000	₹23,58,000	₹1,20,000	360m	1
LP001008	Male	No	0	Graduate	No	₹60,00,000	₹0	₹1,41,000	360m	1
LP001011	Male	Yes	2	Graduate	Yes	₹54,17,000	₹41,96,000	₹2,67,000	360m	1
LP001013	Male	Yes	0	Not Graduate	No	₹23,33,000	₹15,16,000	₹95,000	360m	1
LP001014	Male	Yes	3+	Graduate	No	₹30,36,000	₹25,04,000	₹1,58,000	360m	0
LP001018	Male	Yes	2	Graduate	No	₹40,06,000	₹15,26,000	₹1,68,000	360m	1

Figure 5.5 : Dataset Explorer of AI- Lend Sense: Predictive and Explainable Loan Decision System.

5.6 Admin Dashboard:

In the below screen, we can observe the Admin Dashboard of the AI-Lend Sense system. This dashboard is designed to monitor overall system performance, fairness metrics, and model health. It provides a summary of key statistics such as:

Total Dataset Records – Shows the total number of loan records available in the dataset.

Approval Rate – Displays the percentage of approved loans based on historical data.

Average Loan Amount – Indicates the average loan value across all applicants.

Auto-Approved Rate – Shows the percentage of loans automatically approved within risk limits.

System Status – Displays whether the system is operational.

This section gives administrators a quick overview of the system's current performance.

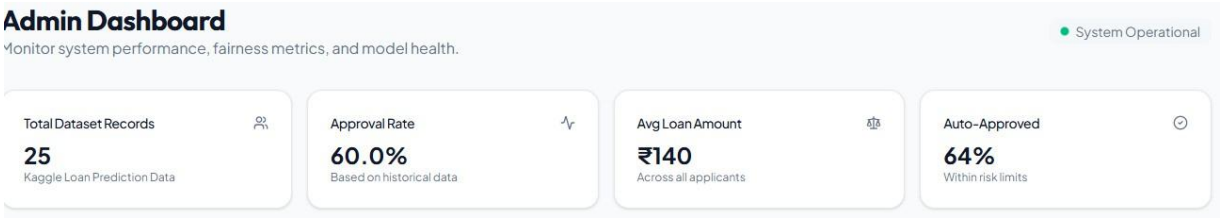

Figure 5.6 : Admin Dashboard of Ai – Lend Sense: Predictive And Explainable Loan Decision System.

5.7 Fairness & Bias Analysis Page:

In below graph, it shows Demographic Parity Check (Gender)

This graph shows approval rates across different genders to ensure there is no unfair discrimination in loan approval decisions.

Loan Amount Distribution

This chart represents the average loan amount approved for each group, helping administrators analyze if any group is receiving disproportionately high or low loan amounts.

This page ensures ethical AI practices and transparency in decision-making.

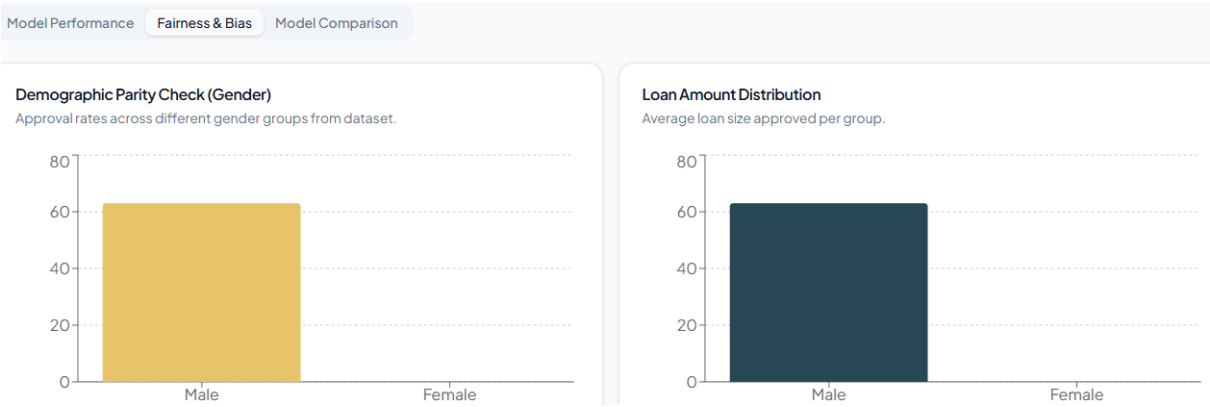

Figure 5.7 : Fairness & Bias Analysis Page of Ai – Lend Sense: Predictive And Explainable Loan Decision System.

5.8 Accuracy vs Fairness Over Time :

This page tracks the trade-off between model accuracy and fairness over a period of time (monthly basis).

The graph shows trends in prediction accuracy.

It also shows fairness scores to ensure ethical compliance.

This helps administrators maintain a balance between high predictive performance and fairness constraints.

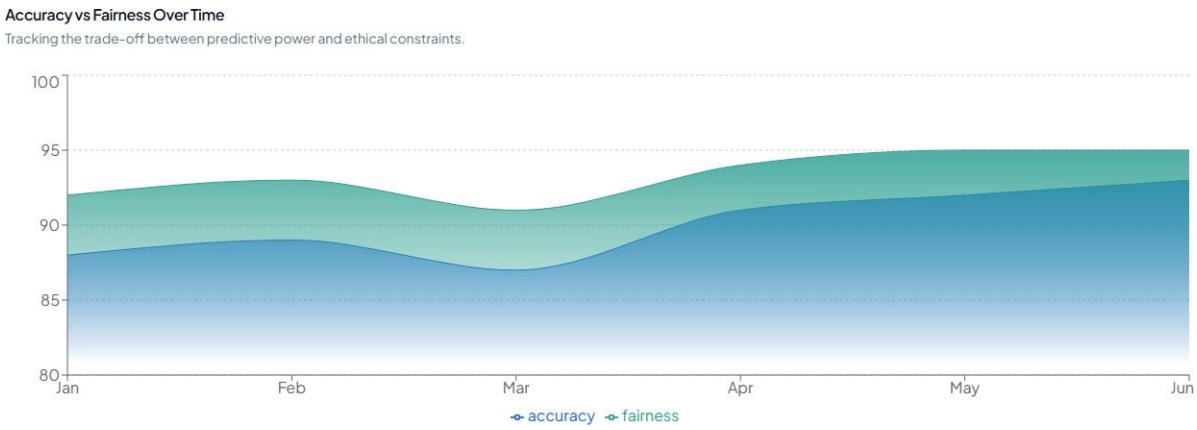

Figure 5.8 : Accuracy vs Fairness Over Time of Ai – Lend Sense: Predictive And Explainable Loan Decision System.

5.9 Model Benchmarking Page:

In below screen, The Model Benchmarking page compares multiple machine learning algorithms used in the system, such as:

Logistic Regression

Decision Tree

Random Forest

The comparison is based on metrics like: → Accuracy

→ Precision

→ Recall

→ F1 Score

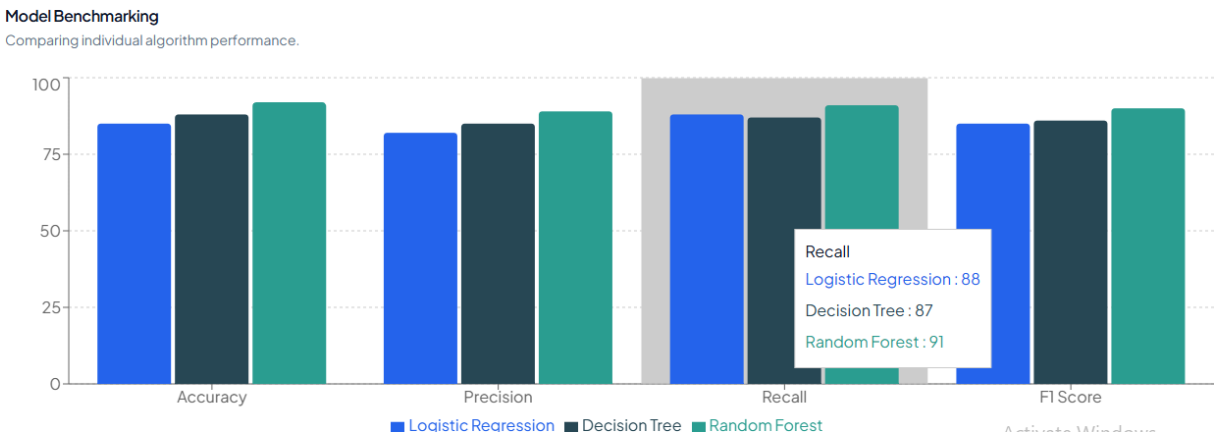

Figure 5.9 : Model Benchmarking Page of Ai – Lend Sense: Predictive And Explainable Loan Decision System.

5.10 Decision Analysis Report:

In the below screen, we can observe the Decision Analysis Report generated by the AI-Lend Sense system after evaluating the loan application. This page displays the final decision of the model along with the associated risk level and confidence score.In this case, the loan application is marked as Rejected (High Risk) based on the ensemble analysis of multiple machine learning models such as Logistic Regression, Decision Tree, and Random Forest. The report also provides a Confidence Score, which indicates the certainty level of the prediction made by the system. Additionally, this page contains a detailed explanation of the key factors that contributed to the rejection decision, such as income level, debt-to-income ratio, credit history, or loan amount risk.This transparent explanation improves trust in the AI system by clearly showing the reasons behind the loan rejection decision

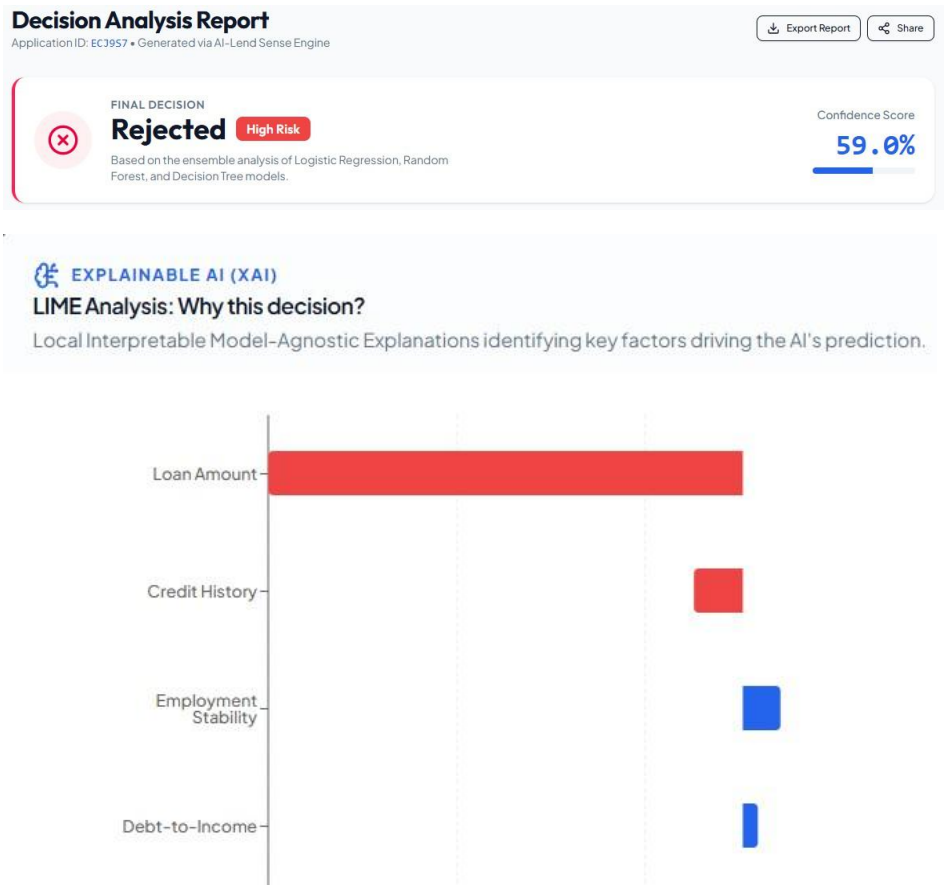

Figure 5.10 : Decision Analysis Report showing Loan Rejection with Risk Level and Confidence Score of Ai – Lend Sense: Predictive And Explainable Loan Decision System.

6. VALIDATION

6. VALIDATION

The validation of this project primarily relies on extensive testing and well-defined test cases to ensure the accuracy and effectiveness of the loan approval prediction system. The testing process involves multiple stages, including dataset validation, model performance evaluation, and real-time prediction testing. By implementing a structured validation approach, we can ensure that the system consistently delivers high accuracy in predicting loan approvals while minimizing errors and misclassifications.

6.1 INTRODUCTION

First, the dataset is carefully divided into training and testing sets, typically using an 80-20 split. The training set is used to train the machine learning models, while the testing set is utilized to evaluate their performance on unseen data. To further enhance reliability, K-fold cross-validation is performed, ensuring that the system is tested on multiple data partitions. This method prevents overfitting and ensures that the model can generalize well to new loan applications.

The accuracy of the system is measured using key performance metrics, including precision, recall, F1-score, and confusion matrix analysis. The confusion matrix provides valuable insights into correct and incorrect predictions, helping refine the model for better results. Additionally, the proposed Explainable AI model integrated with Logistic Regression, Decision Tree, and Random Forest is compared with individual models, demonstrating that the proposed approach achieves better accuracy and transparency.

Finally, real-time prediction testing is conducted by providing new applicant data, ensuring that the system performs effectively in real-world scenarios. Continuous improvements are made based on test results, allowing the model to remain accurate in predicting loan approvals. This structured validation process ensures that the proposed system is reliable, scalable, and capable of maintaining high prediction accuracy in practical applications.

6.2 TEST CASES

TABLE 6.2.1 UPLOADING DATASET

Test case ID	Test case name	Purpose	Test Case	Output
1	User uploads Dataset.	Use it for loan prediction	The user uploads the dataset, on which the loan approval model is trained	Dataset successfully loaded.

TABLE 6.2.2 CLASSIFICATION

Test case ID	Test case name	Purpose	Input	Output
1	Classificatio n test 1	To check if the model predicts correctly.	Eligible applicant details are given	Loan Approved
2	Classificatio n test 2	To check if the model predicts correctly.	Non-eligible applicant details are given	Loan Rejected

7. CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

In conclusion, the project has successfully achieved its objectives, showcasing significant progress and outcomes. The implementation and execution phases were carefully designed and carried out, leading to meaningful improvements in predicting loan approval decisions. Looking ahead, the future aspects of the project hold great potential. Future developments will focus on improving model accuracy, integrating advanced Explainable AI techniques, and enhancing system scalability. These improvements will strengthen the existing system and open new opportunities for innovation, ensuring that the project remains useful and impactful in real-world financial decision-making. This strategic approach will support continuous improvement and long-term success.

7.1 PROJECT CONCLUSION

A robust machine learning framework for loan approval prediction by integrating **Logistic Regression**, **Decision Tree**, and **Random Forest** algorithms along with **Explainable AI (LIME)** for better transparency and interpretability. The model efficiently analyzes applicant data such as income, credit history, loan amount, and other financial factors to predict loan approval with high accuracy. The use of multiple algorithms ensures better performance compared to single-model approaches, while Random Forest improves stability and reduces overfitting.

By incorporating Explainable AI techniques, the system provides clear explanations for each prediction, helping users and financial institutions understand the reasons behind loan approval or rejection. This improves trust and decision-making transparency. The model is optimized for performance and can handle real-time predictions, making it suitable for practical applications in banking and financial services. Its modular design allows easy integration with existing systems and scalability for handling large datasets. Overall, this framework enhances automated loan decision systems by combining machine learning models with interpretability, ensuring reliable and efficient predictions.

7.2 FUTURE ASPECTS

Future aspects include the use of **federated learning**, which enables multiple financial institutions to collaboratively train models without sharing sensitive customer data directly. This can improve model performance while preserving privacy. Another promising direction is the integration of **multimodal data analysis**, where the system can use not only structured financial records but also semi-structured and textual data such as customer feedback, support interactions, or financial documents for better decision-making.

The system can also be enhanced using **graph-based learning techniques**, such as Graph Neural Networks (GNNs), to analyze hidden relationships between applicants, accounts, transactions, and financial entities. This would help in identifying indirect credit risk patterns that traditional models may fail to detect. Another future improvement is the adoption of **transfer learning**, where models pre-trained on large financial datasets can be adapted for loan prediction tasks with limited local data, reducing training time and improving performance.

Further development can include **cross-platform deployment**, allowing the system to be used not only in web applications but also in mobile banking platforms and cloud-based loan management systems. This would increase accessibility for both customers and financial institutions. In addition, the system can be extended to support **multi-loan category prediction**, where it can evaluate different types of loans such as personal loans, home loans, vehicle loans, and education loans separately based on their specific risk factors.

Another valuable enhancement is the implementation of **scenario-based risk simulation**, where the system can predict how changes in applicant conditions, such as salary reduction or increased debt, may affect loan eligibility. This would help banks make more robust lending decisions and support financial planning. Finally, future systems can include **regional and demographic adaptability**, enabling the model to adjust according to local economic conditions, lending policies, and customer financial behavior across different regions or institutions.

8. BIBLIOGRAPHY

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8.2 GITHUB LINK

https://github.com/bhavyabhargavi/LOAN-DECISION-SYSTEM-.git

8.3 PUBLICATIONS

PAPER LINK

https://rdems.in/index.php/RDEMS/article/view/91

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AI-LEND SENSE: PREDICTIVE AND EXPLAINABLE LOAN DECISION SYSTEM

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ABSTRACT

AI-Lend Sense is a loan decision assistant that's easy to understand and use. It helps people figure out loans in a way. Let us be real: banks need loans to make money. It is hard to know who will pay back the loan on time. The old way of checking loans takes a time there is a lot of paperwork and sometimes people do not even know why they were not given a loan. AI-Lend Sense wanted to make this process better. Our tool uses information like a steady income, a stable job, your CIBIL score, how much debt you have compared to your income and verified bank information and the loan amount you want. We put all this information into computer programs like Logistic Regression, Decision Tree and Random Forest. These programs help us make decisions. In a few seconds AI-Lend Sense can tell us if we should say yes or no to a loan. What makes us different is that we are transparent. The people who apply for a loan get a report that is just, for them and this report breaks down the decision that was made about their loan. The lenders can see exactly what is driving each decision so they know that the loan process is fair and based on information. The loan process runs in a clean and easy to use system web app. Borrowers can apply, see insights from the data, and download a detailed analysis of their results. Conversely, the bank managers receive an "admin fee" for each transaction. Based on this, it blends accurate forecasting with simple honesty. Explanations, AI- Lend Sense allows lending to occur faster by enabling AI to more equitable, and even justifiable for the banks and customers alike.

Keywords - Loan Prediction, Logistic Regression, Random Forest, Decision Tree, Explainable AI.

I. INTRODUCTION

The truth is that no one has the time to wait for weeks to even have their questions answered on a loan. However, many banking institutions are also still buried under piles of paperwork and old guts to even deliver a yes or no to the awaiting masses. A yes or no response will only send others on a road to frustration for banking institutions.

The moment you touch down on our Home page, a friendly web portal walks you through step by step. You not only get a chance to test eligibility but also get to see data for your decision or move on for some system-level information. The 'Apply for Loan' form picks up just what it needs—you, your information, your PAN/Aadhaar, your banking information,

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and finally, your financial information—and then tests it out on three good models (Logistic Regression, Decision Tree, Random Forest).

But this is where our service truly sets itself apart. With our explainable AI powered by LIME, we provide a readable explanation for every single decision. We'll start with the decision itself, its confidence level, the risk factor, the votes cast by the individual model, the results of fairness tests, and explanations like "loans are large," "credit scores are low," and "debt burden weighs heavily." We also want to help you increase the chances of a better decision the next time around.

Admins receive a "Dashboard to monitor how models are performing, approval rates, and whether there is any kind of bias creeping in." The "Dataset Explorer allows anyone to view data that drives the entire process." In other words, AI Lend Sense puts accuracy, transparency, and ethics above everything else because AI Lend Sense is one of the few applications of AI that earns trust by weeding out bias to make taking out a loan more characteristic of human behavior.

II. LITERATURE REVIEW

What early loan decisioning systems were trying to fix was the slowness of the manual process. What they replaced it with were simple rule- based decisioning systems. What Thomas and others did was let the experts decide on hard thresholds for things such as income, credit scores, and payments. Well, sure, this moved the process a little quicker, but they were very rigid.

And once more data started pouring in, then statisticians leaped on to Logistic Regression. Hand and Henley demonstrated that you could actually forecast who'd default by looking at income, job status, and credit history. These models were easier to explain than the old rule-based stuff, but they were stuck with linear relationships—and real-world finance is just way messier and more tangled than that.

And then came the tree-based algorithms Breiman's Decision Trees broke down applicants into tidy little boxes governed by rules. These were better, clearer, and more effective than purely statistical methods. But the drawback here is that these algorithms tend to overfit, especially in cases where you lack data or lack balanced data. Then followed ensemble methods like Random Forests, courtesy of Liaw & Wiener. You now had several trees working together, wherein the predictions were more accurate and not shaky. The catch-the models now became "black boxes." Lenders and borrowers couldn't see why a decision landed the way it did.

But then came Explainable AI (XAI), which changed all that. LIME by Ribeiro began to concentrate on which aspects were playing the lead role in arriving at a particular result. Results such as poor credit ratings or excessive debt. People were more comfortable around this process. But to be honest, most of them were still somewhat technical. On top of all that, fairness emerged as a concern (due to Barocas et al.). Automated models to compute credit scores can reinforce gender or geolocation bias. It was proposed that fairness models be used, but essentially, most of these tools simply monitor bias occurrences in the background. Nothing to look at to see where things stand.

Lately, online platforms, for example, one developed by Chen and his teammates, have started integrating data entry with predictive tools and admin interfaces. Yet they also fail to offer decision reports and feedback to the applicants. The problem is that every solution does

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one thing amazingly well but hardly any solution does everything well AI-Lend Sense aims to do in its attempt to fill the solution gap in predicting eligibility, justifying its results, checking for justice in its verdict, allowing you to peep into its data, and providing a report for applicants that's analytically correct. The aim is simple: integrate 'ensemble models, XAI, justice checks, data transparency, and interactive reports in one crystal clear, crystal transparent loan decision solution'.

III. METHODOLOGY

AI-Lend Sense makes getting a loan really simple. It goes through every step of the way and makes sure everything is fair for everyone no matter if you are the one asking for the loan or if you work at the bank. AI- Lend Sense does everything automatically. Tells you why it makes certain decisions. This way AI-Lend Sense gets rid of a lot of the confusion that people usually have when they are trying to get a loan approved by the bank. AI Lend Sense is really good, at explaining things so you know what is going on with your loan. Here is Here is what happens when you want a loan. You go to the website. Click on the button that says Apply for Loan. The system will ask you for some information. They need to know your name. They need to see your government IDs like PAN or Aadhaar. They also need to know about your bank and how money you earn. The system wants to know how long you have been working at your job. They need to know about your debt and how it compares to the money you earn.

They need to know how money you want to borrow and for how long. You also have to tell them why you need the loan. When you give the system all this information it has what it needs to make a decision, about your loan application.

When you send in your details AI-Lend Sense looks over everything to find any mistakes or missing information. It cleans up the data. Gets it ready to use. Then AI-Lend Sense uses machine learning. Your application goes through three models: Logistic Regression, Decision Tree and Random Forest. Each of these models Logistic Regression, Decision Tree and Random Forest tries to make a decision based on what it has learned from applications.

But here is the part: you do not just get a plain yes or no. The system uses AI with LIME to actually show you why it made its decision. Maybe you asked for a loan your credit score is not great you owe a bit too much money already or you have not been in your job for a long time. Whatever the reason the Explainable AI system lays it out in language and with clear visuals so you can understand why the Explainable AI system made its decision, about your loan.

Fairness is very important. The system looks at itself to see if it is being fair. It uses checks to make sure things like gender or where you are from do not affect your results. The system checks for fairness by running checks. If something does not seem right, with the fairness of the system it will catch the problem with the fairness of the system.

At the end you get a Decision Analysis Report. This report tells you if you got the loan or not. It also tells you how sure the system was about its answer. The Decision Analysis Report shows what risk group you are in. It even shows how each model voted on your loan. You will see why you got that result.

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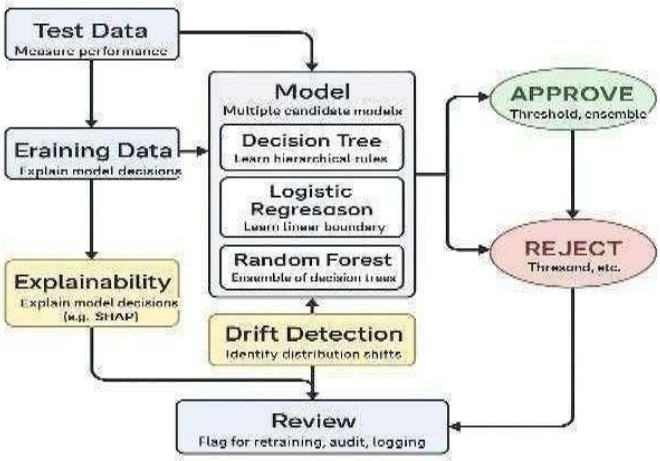

Fig.1. Workflow of the AI-Based Loan Approval System

IV. IMPLEMENTATION

We made -Lend Sense to make loan decisions better and more fair. The system is easy to understand and works in a way. It is made up of parts that work together so it can check if someone is eligible for a loan automatically explain the reason for every decision in a way that is easy to understand and make sure everything is fair. You do not have to do a lot of things to use AI- Lend Sense. You can just open it in your internet browser. Whether you are someone who is applying for a loan from AI-Lend Sense or you are reviewing loan applications or you are in charge of AI-Lend Sense, as an administrator AI-Lend Sense is ready for you to use.

Testing was really easy. The system works well on any computer. The system runs smoothly long as you have Windows 10 or a newer version. We tried the system on our computers and, on the cloud to see how it works in real life. We wanted to make sure the system works well and does not have any problems when many people use it. For the end we chose React, Vite and TypeScript. The front end is quick and responsive. It gives users a lot of things like a Home Page, a Loan Application Form, a Dataset Explorer, an Admin Dashboard and a Decision Analysis Report. When you send in your application the system starts working on your information away. It takes care of missing parts makes sure the numbers are correct and puts things into groups. One important thing it does is calculate your Debt-to- Income ratio. Your Debt-to-Income ratio is found by taking your monthly debt and dividing it by your gross monthly income. When the data is ready it goes to the machine learning models.

We use three machine learning models: Logistic Regression, Decision Tree and Random Forest. The Logistic Regression model combines your information, like your income, your credit score the loan you requested and your job history with some learned weights and a bias. If the Logistic Regression model thinks you have a chance of being approved and this chance is, above a certain level then you are approved by the Logistic Regression model.

If not the Logistic Regression model says no. z

$= w1x1 + w2x2 + \cdots + wnxn + b$

P (Approval) = $1/(1+e^{-z})$

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A Decision Tree looks at the data. Splits it based on the most important thing, about it using something called the Gini Index to tell the good candidates from the bad ones. Random Forest goes one step further. It makes a lot of these trees. Then asks each of them what they think and it decides what to do based on what most of the Decision Trees say.

Final Decision = Majority Vote of All Trees.

We wanted every decision to be clear, so we used Explainable AI (XAI) with LIME. LIME breaks down each prediction, highlighting which details— like a low credit score or a high DTI—moved the needle. You see these right in your Decision Analysis Report, written in plain terms.
It applies fairness constraints such as demographic parity to ensure that the approval rates are not affected by attributes that should not matter.
We looked at the system, with loan cases and fake ones. We tried the system with real and fake loan situations. To find out how good it was we checked how accurate it was, how precise it was, how well it remembered things and its F1-score. The goal was to build a system that's smart, fair and easy to understand so the loan system is smart, fair and easy to understand. These evaluation results show that the system does a lot more than just make predictions. It makes decisions and reliable decisions. The system balances being precise and getting everything through something called the F1-score. This means the model does not send out alerts and it does not miss important cases. This is very important when the system is actually being used in the world. The system also makes sure its decisions are fair. The system was tested with data and fake data to make sure it works well in different situations. The evaluation results show that the system is robust and works with different types of data. The system makes decisions and this is because it uses the evaluation results to make sure it is working correctly. These things show that the way we want to do this is good because it gives the answers and it is fair, to everyone.
True Positive (TP): Disaster correctly identified— alert and allocation triggered.
False Positive (FP): Disaster incorrectly detected .
False Negative (FN): Disaster missed by the system.
True Negative (TN): No disaster, system gets it right Evaluation Metrics.

Precision = $TP/\ TP + FP$

Recall = $TP/\ TP + FN$

F1-Score = 2 * (precision*recall)/(precision+recall)

Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$ (

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V. RESULT AND ANALYSIS

We tried out the AI-Lend Sense system to see how it works with loan applications. We wanted to know if it can tell us who will get a loan and why. These are good results. This system is able to approve a loan without being unjust and complex. We evaluated absolutely everything. We reviewed how people are using this website. We reviewed what the system is predicting. We reviewed what it provides for explanations. We reviewed even the tools administrators are using to monitor what is happening. The AI-Lend Sense system is a good system, doing what it is designed for. It is able to approve a loan. Explain why it is choosing those specific options. Let us begin with the top. The Home Page is where everything starts on a website.

Tone--everything is easy to find. You can jump straight to eligibility criteria, apply, or dig through datasets. Click "Get Started" and you are in the application form.

Fig. 2. Home Page of Loan Approval System.

It is comprehensive, covering a wide array of fields in the loan application form: personal information, government- issued IDs such as PAN or Aadhaar, bank account details, financial information (income levels, debt obligation), and complete details of the loan applied for- amount needed and tenure required.

In case of any incomplete or incorrect information, it gets flagged immediately by the system itself, even before submission. That keeps the data clean and the predictions sharp.

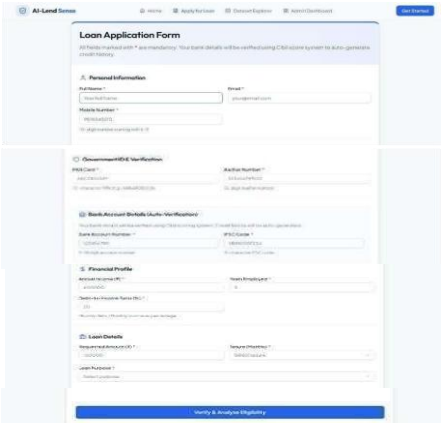

Fig.3. User Loan Application Form For Loan Approval System

When you click on "Verify & Analyze Eligibility" the system starts doing its job. It gives you

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a Loan Decision Analysis Report. This is where things get interesting. The Loan Decision Analysis Report is where you can see what is happening in regards to your loan. Three separate computer models, Logistic Regression & Decision Tree, as well as Random Forest, examine your data. They then make a decision about whether you can have the loan or not. The Loan Decision Analysis Report does not simply provide you an answer yes or no for the loan.
They also provide you with a score indicating how certain the computer is about its decision.

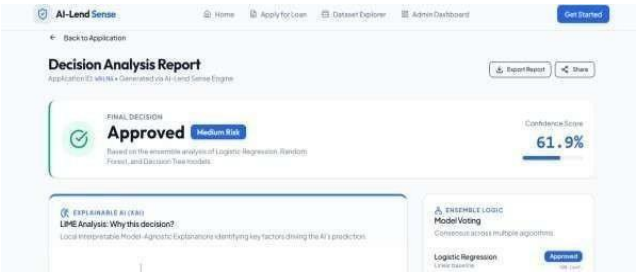

Fig.4. Loan Approval Result Page.

Fig.5. Loan Approval Result Page displaying a Result Description.

"Explanation" really shines in this case. "LIME" allows it to point out what it liked or didn't like about your application, in graphs and easy-to- understand language. If it denied your application, it'll show you what to work on next time. "It increases trust, which trumps algorithms that simply respond "yes" or "no" without an explanation".
If we want to dig deeper, the Dataset Explorer opens up the real data used to train and test the models. You could look at things like income, credit score, loan amount, approval status- there is nothing hidden or shady, so it shows you exactly how the decisions get made.

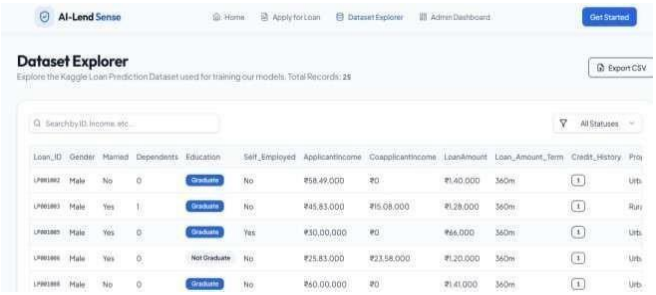

Fig.6. Data Set Exploration of Loan Approval System

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Fig.7. Model Performance Monitoring Dashboard of Loan Approval System.

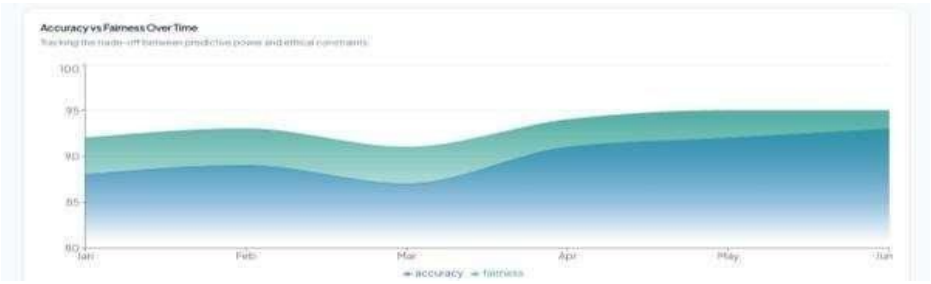

Fig.8. Model Performance Over Time.

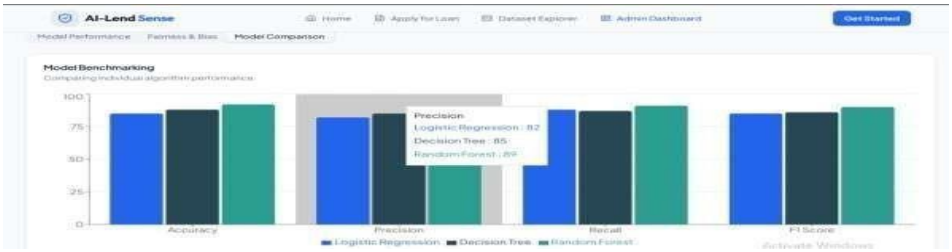

Fig.9. Machine Learning Model Comparison Graph.

V. CONCLUSION

AI-Lend is a completely new way of undertaking loan approvals. That is why it is impressive: bye-bye to lengthy processes and ambiguous approval and disapproval of loans because this is a system employing machine learning and Explainable AI. This takes away the clutter and vagueness surrounding approval and disapprovals of loans. A considerable amount of work is done compared to the former procedure. There is no tiresome loan evaluation process since it acts narrowly.

At the core of AI-Lend Sense are models such as Logistic Regression, Decision Trees, and Random Forests. An individual has come up with a recipe that will not affect predictability with regards to results from risk assessment. In fact, it will not fail even with tough assessments. After the individual has placed a loan request, there will be a compilation of everything that is needed for placing across a loan request. These will include details such as

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personal details, government-approved identification verification, banking details, history, and amounts borrowed from personal loans.

However, the thing that makes this tech even more interesting is LIME, and it really takes the black box out from artificial intelligence. They are not inventing a language; they are illustrating what the most important variables are out of the ones that are involved. The scales of justice had finally been turned in their favor, and they are not sneaky because you can literally see what factors went into a given decision. They are even making sure that it is not prejudiced against any population.

Regarding speed, the AI-Lend Sense outpaces the traditional approach and delivers much better consistency in trialing experience with tools in the possession of the user in the form of the Dataset Explorer and the Admin Dashboard, which enables the user to dig deeper into the numbers, assess how models perform, and monitoring approval rates as well as fairness. AI-Lend Sense's role extends well beyond the fail-forward and futuristic nature of the technology.

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